Yi-Chun Liao

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RESEARCH INTERESTS

Focus on the intersection of computer architecture and machine learning, especially on co-designing hardware and software to enhance resource efficiency and scalability of domain-specific memory-centric accelerators.

EDUCATION

National Taiwan University (NTU) **Taipei, Taiwan**
B.Sc. in Computer Science and Information Engineering (CSIE) *Sep. 2021 – Jan. 2026*

- Overall GPA: 3.90/4.30, Last 60 GPA: 4.09/4.30
- Deferred graduation for 1 semester due to compulsory military service.

PUBLICATIONS

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. "RETENTION: Resource-Efficient Tree-Based Ensemble Model Acceleration with Content-Addressable Memory," Under review by *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025, doi: 10.48550/arXiv.2506.05994

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. "Practicalizing Tree-Based Model Acceleration with CAM through Model Pruning and Data Placement Optimization." *ACM/IEEE International Conference on Hardware/Software Codesign and System Synthesis (CODES+ISSS '25)*, September 28-October 3, 2025, Taipei, Taiwan. ACM, New York, NY, USA, 2 pages. doi:10.1145/3742873.3758342

Hong-Wei Li, Ping-Ting Liu, Bo-Wei Lin, **Yi-Chun Liao**, and Yennun Huang. "IPMES: A Tool for Incremental TTP Detection Over the System Audit Event Stream," *2024 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, Brisbane, Australia, 2024, pp. 265-273, doi:10.1109/DSN58291.2024.00036

RESEARCH EXPERIENCE

Research Intern **Indiana, United States**
University of Notre Dame, Advisor: Prof. Xiaobo Sharon Hu *May 2025 – Present*

- Conduct independent research on realizing in-memory retrieval for RAG, leveraging a quantization scheme to meet hardware constraints and proposing an algorithm to introduce new capability for content-addressable memory (CAM).
- Conduct independent research on alleviating the scalability issue of in-memory hyperdimensional computing, redesigning model to optimize performance under hardware constraints and introducing an algorithm to enhance CAM scalability.
- Contribute to the extension of open-source CAM simulator CAMASim, focusing on data placement optimization with the approaches proposed in the first-authored paper RETENTION.

Research Assistant **Taipei, Taiwan**
Academia Sinica, Advisors: Prof. Tei-Wei Kuo and Prof. Yuan-Hao Chang *Apr. 2024 – Present*

- Conduct independent research on minimizing resource consumption of in-memory inference for tree-based model, introducing an end-to-end framework RETENTION to enhance resource efficiency for CAM-based acceleration.
- Explore and quantify the relationship between capacity redundancy and processing overhead, enabling tradeoffs between controllable accuracy loss, energy consumption, and capacity requirement.
- Propose a pruning algorithm to minimize model complexity within controllable accuracy loss, and two data placement strategies to prioritize space and energy efficiency respectively.
- Improve space efficiency by 4.35× to 207.12× with less than 3% accuracy loss and no degradation on throughput and energy efficiency. The first-authored paper is currently under review by TCAD.
- Collaborate on minimizing retrieved tokens to accelerate RAG, exploring the tradeoffs between accuracy, chunk size, and capacity requirement.

Research Assistant **Taipei, Taiwan**
Academia Sinica, Advisor: Prof. Yennun Huang *Jul. 2023 – Jun. 2024*

- Collaborated on developing intrusion detection tool IPMES to detect incremental tactics, techniques, and procedures (TTPs) in system audit event streams. The co-authored paper was published in DSN 2024.
- Constructed an automated system independently to record hundreds of system attacks, utilizing MITRE Caldera to inject malware and TTPs into virtual machines. The system was subsequently used for synthesizing system attack datasets.

HONORS

First Place of 2025 Bachelor Thesis Award **Taipei, Taiwan**
National Taiwan University *Jun. 2025*

- Distinguished through a three-stage selection process and recognized as one of the top 2 in the CSIE Department, the best in the EECS College, and one of the top 6 university-wide.

Appier Enterprise Award **Taipei, Taiwan**
National Taiwan University *Jun. 2025*

- Received the Most Potential Award from Appier in the departmental undergraduate research exhibition.

Dean's List Award **Taipei, Taiwan**
National Taiwan University *Mar. 2025*

- Ranked 1/155 in the department for both Spring and Fall 2024 semesters.

Irving T. Ho Memorial Scholarship **Taipei, Taiwan**
National Taiwan University *Feb. 2025*

- Awarded to 3 students in the department for academic and research excellence.

First Place of AUO Enterprise Award **Taipei, Taiwan**
MakeNTU 2024 (Nationalwide Makerthon) *May 2024*

- Designed an interactive in-car RPG game for AUO smart windows to enhance travel experience, which was recognized as the best work for AUO Enterprise Award among 43 teams.

TEACHING EXPERIENCE

Teaching Assistant **Taipei, Taiwan**
Computer Architecture, Lecturer: Prof. Chia-Lin Yang *Sep. 2024 – Jan. 2025*

- Supported the instruction of the junior compulsory course Computer Architecture at NTU, enrolled by over 100 students.
- Led weekly discussion sessions, designed and/or graded assignments, quizzes, and exams.
- Provided voluntary office hours, exceeding 10 hours per week during exam periods.

Teaching Assistant **Taipei, Taiwan**
APCS Camp *Jul. 2022/23 – Aug. 2022/23*

- Provided TA hours for over 300 C++ beginners during two-week crash courses.
- Adapted teaching strategies including programming problem solving and customized course reviews to meet the diverse needs and skill levels of students.

EXTRACURRICULAR ACTIVITIES

President **Taipei, Taiwan**
NTU CSIE Department Student Association *Aug. 2023 – Jul. 2024*

- Directed the student council of the CSIE department with over 50 members.
- Promoted student welfare and represented over 600 students in faculty meetings, contributing to departmental decisions.

Vice President **Taipei, Taiwan**
NTU EECS College Student Association *Aug. 2023 – Jul. 2024*

- Represented over 1500 students in college faculty meetings and helped shape student-related policies.

President **Taipei, Taiwan**
NTU Cancer Children Service Club *Feb. 2023 – Jul. 2023*

- Led a 50-member service club, managing budget allocation and arranging educational and recreational activities.
- Worked with Childhood Cancer Foundation of R.O.C., organizing voluntary services to reduce the burden of parents.

Vice Coordinator **Taipei, Taiwan**
NTU CSIE Camp *Feb. 2023 – Jul. 2023*

- Organized a 6-day camp for over 100 high school students with a 50-member staff team.
- Designed activities to introduce students to the CSIE department and provide insights into university life.

SKILLS

- **Language Proficiency: English (Proficient, TOEFL iBT 112/120 with Speaking 28/30)**, Mandarin (Native)
- **Technical Skills:** C, C++, Python (PyTorch, NumPy, scikit-learn), Java, JavaScript, LaTeX, Linux, Git